

Lab Title:- FIREWALL TRAFFIC CONTROL AND FORENSIC VERIFICATION

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Course Name:- BASIC NETWORKING SKILLS FOR DIGITAL FORENSICS

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Executive summary

The lab covers a simple firewall traffic analysis covering some aspect such as blocking/unblocking and also capturing of blocking/unblocking traffic.

Lab Objectives

The objectives of the lab is to understand how firewall traffic control works and how to conduct some basic firewall traffic analysis.

Tools and Resources Used

- Kali linux:- A penetration testing platform used for security assessments.

Kali linux tools

- Apache2:- this is the Apache HyperText Transfer Protocol (HTTP) server program that is designed to be run as a standalone daemon process.
 - Wireshark:- is a GUI network protocol analyzer that lets you interactively browse packet data from a live network or from a previously saved capture file.
 - Iptables:- administration tool for IPv4/IPv6 packet filtering and NAT
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Screenshot and Evidence

SCREENSHOT SHOWING THE ORIGINAL RULESET & HASH VALUE

```
(kali㉿kali)-[~/SBT-DF203-Lab6]
$ sudo iptables-save | tee reports/iptables_before.rules
# Generated by iptables-save v1.4.21 on Thu Aug 14 2025 12:00:00 CEST
*filter
:INPUT ACCEPT [0:0]
:FORWARD ACCEPT [0:0]
:OUTPUT ACCEPT [0:0]
-iptables-save -c
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
num pkts bytes target prot opt in out source destination
Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
num pkts bytes target prot opt in out source destination
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
num pkts bytes target prot opt in out source destination
# iptables-save -c
(kali㉿kali)-[~/SBT-DF203-Lab6]
$ sudo iptables -L -n -v --line-numbers | tee reports/iptables_before.txt
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
num pkts bytes target prot opt in out source destination
Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
num pkts bytes target prot opt in out source destination
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
num pkts bytes target prot opt in out source destination
(kali㉿kali)-[~/SBT-DF203-Lab6]
$ sha256sum reports/iptables_before.rules | tee reports/iptables_before_sha256.txt
e3b0c44298fc1c149afbfb4c8996fb92427ae41e4649b934ca495991b7852b855 reports/iptables_before.rules
(kali㉿kali)-[~/SBT-DF203-Lab6]
```

SCREENSHOT SHOWING NETWORK ADDRESSES & APACHE LISTENER

```
(kali㉿kali)-[~/SBT-DF203-Lab6]
$ ip -br address | tee reports/server_interfaces.txt
lo                UNKNOWN                127.0.0.1/8 ::1/128
eth0              UP                  192.168.0.117/24 fe80::46be:fc38:4b6f:704c/64
lxcbr0            DOWN                 10.0.3.1/24 fc42:5009:ba4b:5ab0::1/64

(kali㉿kali)-[~/SBT-DF203-Lab6]
$ ip route | tee reports/server_routes.txt
default via 192.168.0.1 dev eth0 proto dhcp src 192.168.0.117 metric 100
10.0.3.0/24 dev lxcbr0 proto kernel scope link src 10.0.3.1 linkdown
192.168.0.0/24 dev eth0 proto kernel scope link src 192.168.0.117 metric 100

(kali㉿kali)-[~/SBT-DF203-Lab6]
$ sudo ss -lntp | grep ':80' | tee reports/apache_listener.txt
LISTEN 0      *      *      *      *:80      *:*      users:(("apache2",pid=729,fd=4),("apache2",pid=728,fd=4),("apache2",pid=727,fd=4),("apache2",pid=726,fd=4),("apache2",pid=725,fd=4),("apache2",pid=652,fd=4))

(kali㉿kali)-[~/SBT-DF203-Lab6]
```

SCREENSHOT SHOWING SUCCESSFUL BASELINE CURL

```
(kali@kali)-[~/SBT-DF203-Lab6]
$ curl -v --connect-timeout 5 http://192.168.0.117/firewall_lab.html 2>&1 | tee
baseline_curl.txt bs
* Trying 192.168.0.117:80...
* Established connection to 192.168.0.117 (192.168.0.117 port 80) from 192.168.0.
117 port 53114
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           0      0     0    0         0     0         0      0
0         0     0    0         0     0         0      0
ing HTTP/1.x
> GET /firewall_lab.html HTTP/1.1
> Host: 192.168.0.117
> User-Agent: curl/8.21.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Fri, 18 Sep 2026 20:43:04 GMT
< Server: Apache/2.4.68 (Debian)
< Last-Modified: Fri, 18 Sep 2026 20:38:09 GMT
< ETag: "77-65bc7e0dc42bc"
< Accept-Ranges: bytes
< Content-Length: 119
< Vary: Accept-Encoding
< Content-Type: text/html
<
{ [119 bytes data]
100   119 100   119    0    72077    0
* Connection #0 to host 192.168.0.117:80 left intact
<!DOCTYPE html>
<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: YOUR SERVER N
AME</p></body></html>
(kali@kali)-[~/SBT-DF203-Lab6]
```

SCREENSHOT SHOWING THE SERVER IP ADDRESS USED

```
(root@kali)-[/home/kali/SBT-DF203-Lab6]
# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.0.117 netmask 255.255.255.0 broadcast 192.168.0.255
    inet6 fe80::46be:fc38:4b6f:704c prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:8a:35:d2 txqueuelen 1000 (Ethernet)
    RX packets 1552 bytes 114055 (111.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 258 bytes 30215 (29.5 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 26 bytes 1880 (1.8 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 26 bytes 1880 (1.8 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lxcbr0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 10.0.3.1 netmask 255.255.255.0 broadcast 10.0.3.255
    inet6 fc42:5009:ba4b:5ab0::1 prefixlen 64 scopeid 0x0<global>
    ether 02:00:00:00:00:00 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 10 overruns 0 carrier 0 collisions 0

(root@kali)-[/home/kali/SBT-DF203-Lab6]
#
```

SCREENSHOT SHOWING THE CLIENT IP ADDRESS USED

```
[analyst@secOps ~]$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 192.168.0.106  netmask 255.255.255.0  broadcast 192.168.0.255
    inet6 fe80::a00:27ff:fe55:4407  prefixlen 64  scopeid 0x20<link>
    ether 08:00:27:55:44:07  txqueuelen 1000  (Ethernet)
    RX packets 1702  bytes 122991 (122.9 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 529  bytes 47121 (47.1 KB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    inet6 ::1  prefixlen 128  scopeid 0x10<host>
    loop txqueuelen 1000  (Local Loopback)
    RX packets 564  bytes 45891 (45.8 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 564  bytes 45891 (45.8 KB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

[analyst@secOps ~]$
```

SCREENSHOT SHOWING THE ALLOWED HTTP CAPTURE HANDSHAKE/REQUEST/RESPONSE

```
(root@kali)-[/home/kali/GBT-DF203-Lab6]
# sudo tshark -i eth0 -f 'host 192.168.0.106 and tcp port 80' -a duration:30 >
evidence/http_allowed.pcapng &
sleep 3
PCAP in evidence/http_allowed.pcapng evidence/http_blocked.pcapng: done
[1] 12992
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
10
ip.dst == tcp.dstport
done | tee reports/allowed_vs_blocked.tsv
[1] + done sudo tshark -i eth0 -f 'host 192.168.0.106 and tcp port 80' -a
duration:30 >
(root@kali)-[/home/kali/GBT-DF203-Lab6]
# sha256sum evidence/http_allowed.pcapng | tee reports/http_allowed_sha256.txt
bcbe1400aaaa351a840dc069cda3052be0674913d63a777d6a32382e05d46ef0 evidence/http_a
llowed.pcapng
(root@kali)-[/home/kali/GBT-DF203-Lab6]
```

SCREENSHOT SHOWING THE INSERTED DROP RULE AND VERIFICATION

```
(root@kali)-[/home/kali/SBT-DF203-Lab6]
# sudo iptables -I INPUT 1 -s "192.168.0.106" -p tcp --dport 80 -j DROP

(root@kali)-[/home/kali/SBT-DF203-Lab6]
# sudo iptables -L INPUT -n -v --line-numbers | tee reports/iptables_after_add.txt
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
num pkts bytes target prot opt in out source destination
1 0 0 DROP tcp -- * * -- 192.168.0.106 tcp -- 0.0.0.0/0.0.0.0
0 0 0 DROP tcp dpt:80

(root@kali)-[/home/kali/SBT-DF203-Lab6]
# sudo iptables -C INPUT -s "192.168.0.106" -p tcp --dport 80 -j DROP

(root@kali)-[/home/kali/SBT-DF203-Lab6]
# printf 'Rule verified present.\n' | tee reports/rule_verification.txt
Rule verified present.
```

SCREENSHOT SHOWING THE BLOCKED CURL TIMEOUT

```
[analyst@secOps ~]$ curl -v --connect-timeout 10 http://192.168.0.117/firewall_lab.html
* Trying 192.168.0.117:80...
* After 10000ms connect time, move on!
* connect to 192.168.0.117 port 80 failed: Connection timed out
* Connection timeout after 10002 ms
* Closing connection 0
curl: (28) Connection timeout after 10002 ms
```

SCREENSHOT SHOWING THE BLOCKED CAPTURE SHOWING RETRANSMISSIONS/NO RESPONSE

```

└─# sudo tshark -i eth0 -f 'host 192.168.0.106 and tcp port 80' -a duration:35 >
evidence/http_blocked.pcapng &
sleep 3
└─# mv evidence/http_allowed.pcapng evidence/http_blocked.pcapng; do
[1] 17841
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'

└─(root@kali)-[/home/kali/SBT-DF203-Lab6]
4 # tcp.flags > tcp.analysis.retransmission > http.request.uri > http.respo
[1] + done reportsudo tshark -i eth0 -f 'host 192.168.0.106 and tcp port 80' -a
duration:35 >
└─(root@kali)-[/home/kali/SBT-DF203-Lab6]
# sudo iptables -L INPUT -n -v --line-numbers | tee reports/iptables_after_test
.txt
Chain INPUT (policy ACCEPT 8 packets, 328 bytes)
num pkts bytes target prot opt in out source destination
1 6 360 DROP tcp -- * * 192.168.0.106 0.0.0.0/
0 tcp dpt:80
└─(root@kali)-[/home/kali/SBT-DF203-Lab6]
# sha256sum evidence/http_blocked.pcapng | tee reports/http_blocked_sha256.txt
e128a68ae6a08c87401b60aae3b11a8e18ade3cd1003bc77086be4ef7256b849 evidence/http_b
locked.pcapng
└─(root@kali)-[/home/kali/SBT-DF203-Lab6]

```

SCREENSHOT SHOWING THE ALLOWED-VERSUS-BLOCKED COMPARISON

```
└─$ cd SBT-DF203-Lab6
(kali@kali)~[~/SBT-DF203-Lab6]
$ for PCAP in evidence/http_allowed.pcapng evidence/http_blocked.pcapng; do
    echo "==== $PCAP ===="
    tshark -r "$PCAP" -Y 'tcp.flags.syn==1 || http.request || http.response || \
tcp.analysis.retransmission' -T fields -e frame.number -e frame.time_relative -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport \
-e ip.dst -e tcp.dstport \
-e tcp.flags -e tcp.analysis.retransmission -e http.request.uri -e http.response.code \
tee reports/allowed_vs_blocked.tsv
==== evidence/http_allowed.pcapng ====
==== evidence/http_blocked.pcapng ====
(kali@kali)~[~/SBT-DF203-Lab6]
$ cd reports
(kali@kali)~[~/SBT-DF203-Lab6/reports]
$ ls
allowed_vs_blocked.tsv  http_blocked_sha256.txt  iptables_after_test.txt  http_allowed_sha256.txt  iptables_after_add.txt  rule_verification.txt
```

SCREENSHOT SHOWING THE RULE REMOVAL AND RESTORED ACCESS

```
(root@kali)~[~/home/kali/SBT-DF203-Lab6]
# sudo iptables -D INPUT -s "192.168.0.106" -p tcp --dport 80 -j DROP

(kali@kali)~[~/home/kali/SBT-DF203-Lab6]
# sudo iptables -C INPUT -s "192.168.0.106" -p tcp --dport 80 -j DROP || echo '
Rule successfully removed'
iptables: Bad rule (does a matching rule exist in that chain?).
Rule successfully removed

(kali@kali)~[~/home/kali/SBT-DF203-Lab6]
# sudo iptables -L INPUT -n -v --line-numbers | tee reports/iptables_restored.txt
Chain INPUT (policy ACCEPT 10 packets, 392 bytes)
num  pkts bytes target    prot opt in     out     source                 destination
1
(kali@kali)~[~/home/kali/SBT-DF203-Lab6]
#

[analyst@secOps ~]$ curl -v --connect-timeout 5 http://192.168.0.117/firewall_lab.html
* Trying 192.168.0.117:80...
* Connected to 192.168.0.117 (192.168.0.117) port 80 (#0)
> GET /firewall_lab.html HTTP/1.1
> Host: 192.168.0.117
> User-Agent: curl/7.81.0
> Accept: */*
>
* Mark bundle as not supporting multiuse
< HTTP/1.1 200 OK
< Date: Fri, 18 Sep 2026 21:10:27 GMT
< Server: Apache/2.4.68 (Debian)
< Last-Modified: Fri, 18 Sep 2026 20:38:09 GMT
< ETag: "77-65bc7e0dc42bc"
< Accept-Ranges: bytes
< Content-Length: 119
< Vary: Accept-Encoding
< Content-Type: text/html
<
<!DOCTYPE html>
<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: YOUR SERVER NAME</p></body></html>
* Connection #0 to host 192.168.0.117 left intact
[analyst@secOps ~]$
```

Conclusion

In conclusion the lab has help me in understanding the basics of firewall traffic control.

Reference

➤ICDFA course Material (SBT-DF203(basic networking skills for digital forensics)- by
COMMANDANT AMINU IDRIS

➤ICDFA Recorded sessions
